

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(f)	(i)		[H ⁺] = 0.02 (1) pH = 1.7 (1) award (1) for pH = 2		2		2	2	
		(ii)		oxidation state Mn changes from IV to II therefore it is reduced (1) oxidation state Cl changes from –1 to 0 therefore it is oxidised (1)		2		2		
				Question 9 total	0	12	3	15	7	8